

Programming Languages Homework 3 Answer Key

1. C

```
#include <stdio.h>

int main()
{
    x=15;

    int x;

    x=30;

    printf("value of x: %d ", x);

    return 0;
}
```

Result:

error: `x' undeclared (first use this function)

Execution terminated

Java

```
public class test {
    public static void main(String[] args) {
        // TODO code application logic here
        x=15;
        int x;
        x=30;
        System.out.println("value of x =" +x);
    }
}
```

Result:

cannot find symbol: variable x

x=15;

C++

```
#include <iostream>

using namespace std;

int main()

{

    x=15;

    int x;

    x=30;

    cout <<" value of x = " << x << endl;

    getchar();

    return 0;

}
```

Result:

error: 'x' undeclared (first use this function)
Execution terminated

C#

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApplication1
{
    class Program
    {
        static void Main(string[] args)
        {
            x=15;
            int x;
            x=30;
            Console.WriteLine("value of x =" +x);
        }
    }
}
```

Result:

Error:Cannot use local variable 'x' before it is declared.

At the compile-time of the above programs errors have occurred and they haven't performed the work due to the uses of variable x before it is performed as explicit declaration,.

2. Java

```
public class test2 {  
  
    public static void main(String[] args) {  
  
        int[] dizi;  
  
        dizi = new int[10];  
  
        for(int i = 0; i < dizi.length; i++)  
  
        {  
  
            dizi[i]=i;  
  
        }  
  
        for(int i=0;i<=dizi.length;i++)  
  
        {  
  
            System.out.println(dizi[i]);  
  
        }  
  
    }  
  
}
```

Result:

0
1
2
3
4
5

6

7

8

9

Exception in thread "main" java.lang.ArrayIndexOutOfBoundsException: 10

C

```
#include <stdlib.h>
```

```
#include<stdio.h>
```

```
main(){
```

```
    int i,x[10];
```

```
    for(i=10;i>=0;i--)
```

```
    {
```

```
        x[i]=i;
```

```
        printf("%d\n",x[i]);
```

```
    }
```

```
    printf("%d\n",x[-1]);
```

```
    getchar();
```

```
}
```

Result:

10

9

8

7

6

5

4

3
2
1
0

16 (value which is in memory adress that is prior to the beginning of array x (($\&x[0]$ -sizeof(int)))

C++

```
#include <iostream>

using namespace std;

int x[10];

main()
{
    for(int i=10;i>=0;i--){

        x[i]=i;

    }

    cout << " value:" << ::x[-1] << endl;

    getchar();

}
```

Result:

value:0 (if no assignment, default value is equal to zero "0")

Perl

```
#!/usr/bin/perl

@ages = (25, 30, 40);
@names = ("John Paul", "Lisa", "Kumar");

print "\$ages[0] = $ages[0]\n";
print "\$ages[1] = $ages[1]\n";
print "\$ages[5] = $ages[5]\n";
print "\$names[0] = $names[0]\n";
print "\$names[1] = $names[1]\n";
```

```
print "\\$names[-5] = $names[-5]\\n";
```

Result:

\$ages[0]=25

\$ages[1]=30

\$ages[5]=

\$names[0]=John Paul

\$names[1]=Lisa

\$names[-5]=

C#

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApplication1
{
    class Program
    {
        static void Main(string[] args)
        {
            int[] dizi;
            dizi = new int[10];
            for(int i = 0; i < dizi.Length; i++)
            {
                dizi[i]=i;
            }
            for(int i=0;i<=dizi.Length;i++)
            {
                Console.WriteLine(dizi[i]);
            }
        }
    }
}
```

Result:

0

1

2

3
4
5
6
7
8
9

An unhandled exception of type 'System.IndexOutOfRangeException'
occurred for index 10

As can be seen from the results of the above program; while C,C++ ve Perl programming languages do not perform range control for indices of array, Java ve C# programming languages do.